ALLIANCE

UNIVERSITY

Private University established in Karnataka State by Act No. 34 of year 2010
Recognised by the University Grants Commission (UGC), New Delhi

DATABASE MANAGEMENT SYSTEM

CAB BOOKING MANAGEMENT SYSTEM

ASSIGNMENT 7

SAMPLE DATA INSERTION

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Subject: Database Management System (DBMS)

1. Introduction

This assignment inserts realistic sample data into all eight tables of the Cab Booking Management System. The data represents a small cab service operating in Bengaluru, with customers living in areas such as Anekal, HSR Layout, Koramangala and Whitefield, and trips between places such as Alliance University, Electronic City, the airport and MG Road.

The data covers the complete booking lifecycle: completed trips with payments and ratings, cancelled bookings, confirmed bookings, a trip in progress, a pending booking, a pending payment and completed rides that were not rated. This makes it possible to test the constraints and relationships with realistic situations.

2. Insertion Order and Record Counts

Because of the foreign keys, parent tables must be filled before the child tables. Each INSERT lists the column names explicitly so that the values cannot be placed in the wrong column.

Order	Table	Records	Depends on
1	CUSTOMER	10	-
2	DRIVER	8	-
3	CAB	8	DRIVER
4	BOOKING	15	CUSTOMER, DRIVER, CAB
5	TRIP	12	BOOKING
6	PAYMENT	10	BOOKING
7	RATING	8	BOOKING
8	DRIVER_LOCATION	16	DRIVER

Total: 87 records.

3. Rules Followed by the Data

Rule	How the sample data follows it
Primary keys	Every ID is unique: customers 101-110, drivers 201-208, cabs 301-308, bookings 5001-5015, trips 7001-7012, payments 8001-8010, ratings 9001-9008, locations 1-16.
Foreign keys	Every Customer_ID, Driver_ID, Cab_ID and Booking_ID used in a child table already exists in its parent table.
Driver-Cab 1:1	Each driver has exactly one cab (driver 201 has cab 301, and so on) and each booking uses the cab of its driver.
UNIQUE	All phone numbers, emails, licence numbers and vehicle numbers are different. Each booking has at most one trip, one payment and one rating.
CHECK	Statuses and methods use only the allowed values, ratings are 1 to 5, fares and distances are not negative, and every trip ends after it starts.
NOT NULL / NULL	Mandatory fields are always filled. NULL is used only for optional data: one customer email, a payment date for a pending payment, and the end time, distance and fare of a trip still in progress.

4. INSERT Statements

4.1 CUSTOMER

Ten passengers with unique phone numbers and emails. Customer 107 has no email, which is allowed because Email is optional.

```
INSERT INTO CUSTOMER (Customer_ID, Name, Phone, Email, Address) VALUES
(101, 'Rahul Sharma', '9886012101', 'rahul.sharma@gmail.com', 'Chandapura-Anekal Road, Anekal, Bengaluru'),
(102, 'Ananya Iyer', '9886012102', 'ananya.iyer@gmail.com', '24, 5th Cross, HSR Layout, Bengaluru'),
(103, 'Karthik Nair', '9886012103', 'karthik.nair@yahoo.com', '8, 80 Feet Road, Koramangala, Bengaluru'),
(104, 'Sneha Patil', '9886012104', 'sneha.patil@gmail.com', '45, 9th Main, Jayanagar, Bengaluru'),
(105, 'Arjun Mehta', '9886012105', 'arjun.mehta@outlook.com', '16, 100 Feet Road, Indiranagar, Bengaluru'),
(106, 'Divya Krishnan', '9886012106', 'divya.krishnan@gmail.com', 'Phase 2, Electronic City, Bengaluru'),
(107, 'Rohan Das', '9886012107', NULL, 'Anekal Main Road, Anekal, Bengaluru'),
(108, 'Meera Joshi', '9886012108', 'meera.joshi@gmail.com', '32, 2nd Stage, BTM Layout, Bengaluru'),
(109, 'Vikram Singh', '9886012109', 'vikram.singh@gmail.com', 'Attibele Road, Anekal, Bengaluru'),
(110, 'Pooja Reddy', '9886012110', 'pooja.reddy@gmail.com', 'ITPL Main Road, Whitefield, Bengaluru')
```

4.2 DRIVER

Eight drivers with unique phone and licence numbers. Their status is Available, Busy or Offline.

```
INSERT INTO DRIVER (Driver_ID, Name, Phone, License_Number, Status) VALUES
(201, 'Ramesh Kumar', '9845012301', 'KA0120190012345', 'Available'),
(202, 'Suresh Babu', '9845012302', 'KA0520180023456', 'Busy'),
(203, 'Manjunath Gowda', '9845012303', 'KA0320200034567', 'Available'),
(204, 'Imran Khan', '9845012304', 'KA0220170045678', 'Offline'),
(205, 'Venkatesh Rao', '9845012305', 'KA0420190056789', 'Available'),
(206, 'Anil Reddy', '9845012306', 'KA0120210067890', 'Busy'),
(207, 'Prakash Shetty', '9845012307', 'KA0520160078901', 'Available'),
(208, 'Mohammed Irfan', '9845012308', 'KA0320220089012', 'Available')
```

4.3 CAB

Eight cabs, one per driver, of type Sedan, SUV or Hatchback. Cab 304 is under Maintenance and cabs 302 and 306 are currently Booked.

```
INSERT INTO CAB (Cab_ID, Driver_ID, Vehicle_Number, Cab_Type, Model, Status) VALUES
(301, 201, 'KA01AB1234', 'Sedan', 'Maruti Swift Dzire', 'Available'),
(302, 202, 'KA05MN4521', 'SUV', 'Toyota Innova Crysta', 'Booked'),
(303, 203, 'KA03CD7788', 'Hatchback', 'Maruti WagonR', 'Available'),
(304, 204, 'KA02EF3345', 'Sedan', 'Hyundai Aura', 'Maintenance'),
(305, 205, 'KA04GH9021', 'SUV', 'Hyundai Creta', 'Available'),
(306, 206, 'KA01JK5567', 'Hatchback', 'Tata Tiago', 'Booked'),
(307, 207, 'KA05LM1290', 'Sedan', 'Honda Amaze', 'Available'),
(308, 208, 'KA03PQ6643', 'SUV', 'Maruti Ertiga', 'Available')
```

4.4 BOOKING

Fifteen bookings from 1 to 24 October 2026: 10 Completed, 2 Cancelled, 2 Confirmed and 1 Pending. Customers 101, 102, 104, 105 and 107 booked more than once, showing the 1:M relationship.

```
INSERT INTO BOOKING (Booking_ID, Customer_ID, Driver_ID, Cab_ID, Booking_Date, Pickup_Location, Drop_Location,
    Booking_Status) VALUES
(5001, 101, 205, 305, '2026-10-01 08:15:00', 'Alliance University', 'Electronic City Phase 1', 'Completed'),
(5002, 102, 201, 301, '2026-10-02 09:30:00', 'HSR Layout', 'Kempegowda International Airport', 'Completed'),
(5003, 103, 203, 303, '2026-10-03 18:45:00', 'Koramangala', 'Whitefield', 'Completed'),
(5004, 104, 207, 307, '2026-10-04 07:50:00', 'Jayanagar', 'MG Road', 'Completed'),
(5005, 105, 208, 308, '2026-10-06 21:10:00', 'Indiranagar', 'Marathahalli', 'Completed'),
(5006, 106, 201, 301, '2026-10-08 13:20:00', 'Electronic City Phase 2', 'Silk Board Junction', 'Completed'),
(5007, 107, 205, 305, '2026-10-10 10:05:00', 'Alliance University', 'Bannerghatta Road', 'Completed'),
(5008, 108, 203, 303, '2026-10-12 17:40:00', 'BTM Layout', 'Majestic Bus Stand', 'Completed'),
(5009, 109, 207, 307, '2026-10-15 06:30:00', 'Anekal Town', 'Kempegowda International Airport', 'Completed'),
(5010, 110, 208, 308, '2026-10-17 19:25:00', 'Whitefield', 'Indiranagar', 'Completed'),
(5011, 101, 203, 303, '2026-10-19 11:00:00', 'Electronic City Phase 1', 'Alliance University', 'Cancelled'),
(5012, 104, 201, 301, '2026-10-20 15:30:00', 'MG Road', 'HSR Layout', 'Cancelled'),
(5013, 102, 202, 302, '2026-10-23 08:00:00', 'Koramangala', 'Electronic City Phase 1', 'Confirmed'),
(5014, 105, 206, 306, '2026-10-23 09:15:00', 'Marathahalli', 'Jayanagar', 'Confirmed'),
(5015, 107, 205, 305, '2026-10-24 10:30:00', 'Alliance University', 'Silk Board Junction', 'Pending')
```

4.5 TRIP

Twelve trips: 10 completed journeys with distance, fare and times; trip 7011 is Started (no end time or fare yet); trip 7012 is Cancelled. Bookings that were never started have no trip record yet.

```
INSERT INTO TRIP (Trip_ID, Booking_ID, Start_Time, End_Time, Distance, Fare, Trip_Status) VALUES
(7001, 5001, '2026-10-01 08:25:00', '2026-10-01 09:02:00', 18.50, 350.00, 'Completed'),
(7002, 5002, '2026-10-02 09:40:00', '2026-10-02 10:35:00', 36.20, 780.00, 'Completed'),
(7003, 5003, '2026-10-03 19:00:00', '2026-10-03 20:05:00', 22.80, 520.00, 'Completed'),
(7004, 5004, '2026-10-04 08:00:00', '2026-10-04 08:30:00', 9.40, 210.00, 'Completed'),
(7005, 5005, '2026-10-06 21:20:00', '2026-10-06 21:58:00', 14.60, 290.00, 'Completed'),
(7006, 5006, '2026-10-08 13:30:00', '2026-10-08 13:52:00', 8.10, 180.00, 'Completed'),
(7007, 5007, '2026-10-10 10:15:00', '2026-10-10 11:05:00', 21.00, 420.00, 'Completed'),
(7008, 5008, '2026-10-12 17:55:00', '2026-10-12 18:50:00', 12.70, 260.00, 'Completed'),
(7009, 5009, '2026-10-15 06:40:00', '2026-10-15 07:55:00', 41.50, 950.00, 'Completed'),
(7010, 5010, '2026-10-17 19:35:00', '2026-10-17 20:25:00', 17.90, 360.00, 'Completed'),
(7011, 5013, '2026-10-23 08:10:00', NULL, NULL, NULL, 'Started'),
(7012, 5012, NULL, NULL, 0.00, 0.00, 'Cancelled')
```

4.6 PAYMENT

Ten payments using UPI, Card and Cash. Each amount equals the fare of the trip. Payment 8010 is Pending, so its date is NULL.

```
INSERT INTO PAYMENT (Payment_ID, Booking_ID, Amount, Payment_Method, Payment_Date, Payment_Status) VALUES
(8001, 5001, 350.00, 'UPI', '2026-10-01 09:05:00', 'Successful'),
(8002, 5002, 780.00, 'Card', '2026-10-02 10:40:00', 'Successful'),
(8003, 5003, 520.00, 'UPI', '2026-10-03 20:10:00', 'Successful'),
(8004, 5004, 210.00, 'Cash', '2026-10-04 08:35:00', 'Successful'),
(8005, 5005, 290.00, 'UPI', '2026-10-06 22:00:00', 'Successful'),
(8006, 5006, 180.00, 'Cash', '2026-10-08 13:55:00', 'Successful'),
(8007, 5007, 420.00, 'Card', '2026-10-10 11:10:00', 'Successful'),
(8008, 5008, 260.00, 'UPI', '2026-10-12 18:55:00', 'Successful'),
(8009, 5009, 950.00, 'Card', '2026-10-15 08:00:00', 'Successful'),
(8010, 5010, 360.00, 'UPI', NULL, 'Pending')
```

4.7 RATING

Eight ratings between 3 and 5. Bookings 5008 and 5010 are completed but not rated, showing the optional 1:0..1 relationship.

```
INSERT INTO RATING (Rating_ID, Booking_ID, Rating, Review) VALUES
(9001, 5001, 5, 'Good ride, smooth and punctual.'),
(9002, 5002, 4, 'Reached the airport on time, AC could be cooler.'),
(9003, 5003, 5, 'Very polite driver and clean cab.'),
(9004, 5004, 3, 'Driver took a longer route.'),
(9005, 5005, 4, 'Safe driving and a good experience.'),
(9006, 5006, 5, 'Quick pickup and a comfortable ride.'),
(9007, 5007, 4, 'Good ride overall.'),
(9008, 5009, 5, 'Early morning pickup was on time.')
```

4.8 DRIVER_LOCATION

Sixteen location updates, two for each driver, using real Bengaluru coordinates. This shows the 1:M relationship from DRIVER.

```
INSERT INTO DRIVER_LOCATION (Location_ID, Driver_ID, Latitude, Longitude, Updated_Time) VALUES
(1, 201, 12.916500, 77.610100, '2026-10-24 09:00:00'),
(2, 201, 12.925000, 77.615000, '2026-10-24 09:30:00'),
(3, 202, 12.845600, 77.660300, '2026-10-24 09:00:00'),
(4, 202, 12.850000, 77.665000, '2026-10-24 09:30:00'),
(5, 203, 12.935200, 77.624500, '2026-10-24 09:00:00'),
(6, 203, 12.940000, 77.630000, '2026-10-24 09:30:00'),
(7, 204, 12.969800, 77.750000, '2026-10-24 09:00:00'),
(8, 204, 12.970000, 77.751000, '2026-10-24 09:30:00'),
(9, 205, 12.710700, 77.701600, '2026-10-24 09:00:00'),
(10, 205, 12.716000, 77.710000, '2026-10-24 09:30:00'),
(11, 206, 12.956900, 77.701100, '2026-10-24 09:00:00'),
(12, 206, 12.960000, 77.705000, '2026-10-24 09:30:00'),
(13, 207, 12.925000, 77.593800, '2026-10-24 09:00:00'),
(14, 207, 12.930000, 77.599000, '2026-10-24 09:30:00'),
(15, 208, 12.978400, 77.640800, '2026-10-24 09:00:00'),
(16, 208, 12.981000, 77.644000, '2026-10-24 09:30:00')
```

5. Verifying the Inserted Data

All INSERT statements ran without any constraint error. The following queries confirm the data that is stored.

5.1 Record count of each table

```
mysql> SELECT 'CUSTOMER' AS Table_Name, COUNT(*) AS Total_Records FROM CUSTOMER UNION ALL SELECT 'DRIVER',
    -> COUNT(*) FROM DRIVER UNION ALL SELECT 'CAB', COUNT(*) FROM CAB UNION ALL SELECT 'BOOKING',
    -> COUNT(*) FROM BOOKING UNION ALL SELECT 'TRIP', COUNT(*) FROM TRIP UNION ALL SELECT 'PAYMENT',
    -> COUNT(*) FROM PAYMENT UNION ALL SELECT 'RATING', COUNT(*) FROM RATING UNION ALL SELECT
    -> 'DRIVER_LOCATION', COUNT(*) FROM DRIVER_LOCATION;

+-----------------+---------------+
| Table_Name      | Total_Records |
+-----------------+---------------+
| CUSTOMER        |            10 |
| DRIVER          |             8 |
| CAB             |             8 |
| BOOKING         |            15 |
| TRIP            |            12 |
| PAYMENT         |            10 |
| RATING          |             8 |
| DRIVER_LOCATION |            16 |
+-----------------+---------------+
```

5.2 Contents of CUSTOMER

```
mysql> SELECT * FROM CUSTOMER;

+-------------+----------------+------------+--------------------------+-----------------------------------------+
| Customer_ID | Name           | Phone      | Email                    | Address                                 |
+-------------+----------------+------------+--------------------------+-----------------------------------------+
|         101 | Rahul Sharma   | 9886012101 | rahul.sharma@gmail.com   | Chandapura-Anekal Road, Anekal, Bengaluru |
|         102 | Ananya Iyer    | 9886012102 | ananya.iyer@gmail.com    | 24, 5th Cross, HSR Layout, Bengaluru      |
|         103 | Karthik Nair   | 9886012103 | karthik.nair@yahoo.com   | 8, 80 Feet Road, Koramangala, Bengaluru   |
|         104 | Sneha Patil    | 9886012104 | sneha.patil@gmail.com    | 45, 9th Main, Jayanagar, Bengaluru        |
|         105 | Arjun Mehta    | 9886012105 | arjun.mehta@outlook.com  | 16, 100 Feet Road, Indiranagar, Bengaluru |
|         106 | Divya Krishnan | 9886012106 | divya.krishnan@gmail.com | Phase 2, Electronic City, Bengaluru       |
|         107 | Rohan Das      | 9886012107 | NULL                     | Anekal Main Road, Anekal, Bengaluru       |
|         108 | Meera Joshi    | 9886012108 | meera.joshi@gmail.com    | 32, 2nd Stage, BTM Layout, Bengaluru      |
|         109 | Vikram Singh   | 9886012109 | vikram.singh@gmail.com   | Attibele Road, Anekal, Bengaluru          |
|         110 | Pooja Reddy    | 9886012110 | pooja.reddy@gmail.com    | ITPL Main Road, Whitefield, Bengaluru     |
+-------------+----------------+------------+--------------------------+-----------------------------------------+
```

5.3 Contents of DRIVER

```
mysql> SELECT * FROM DRIVER;
```

```
+-----------+-----------------+------------+-----------------+-----------+
| Driver_ID | Name            | Phone      | License_Number  | Status    |
+-----------+-----------------+------------+-----------------+-----------+
|       201 | Ramesh Kumar    | 9845012301 | KA0120190012345 | Available |
|       202 | Suresh Babu     | 9845012302 | KA0520180023456 | Busy      |
|       203 | Manjunath Gowda | 9845012303 | KA0320200034567 | Available |
|       204 | Imran Khan      | 9845012304 | KA0220170045678 | Offline   |
|       205 | Venkatesh Rao   | 9845012305 | KA0420190056789 | Available |
|       206 | Anil Reddy      | 9845012306 | KA0120210067890 | Busy      |
|       207 | Prakash Shetty  | 9845012307 | KA0520160078901 | Available |
|       208 | Mohammed Irfan  | 9845012308 | KA0320220089012 | Available |
+-----------+-----------------+------------+-----------------+-----------+
```

5.4 Contents of CAB

```
mysql> SELECT * FROM CAB;
```

```
+--------+-----------+----------------+-----------+----------------------+-------------+
| Cab_ID | Driver_ID | Vehicle_Number | Cab_Type  | Model                | Status      |
+--------+-----------+----------------+-----------+----------------------+-------------+
|    301 |       201 | KA01AB1234     | Sedan     | Maruti Swift Dzire   | Available   |
|    302 |       202 | KA05MN4521     | SUV       | Toyota Innova Crysta | Booked      |
|    303 |       203 | KA03CD7788     | Hatchback | Maruti WagonR        | Available   |
|    304 |       204 | KA02EF3345     | Sedan     | Hyundai Aura         | Maintenance |
|    305 |       205 | KA04GH9021     | SUV       | Hyundai Creta        | Available   |
|    306 |       206 | KA01JK5567     | Hatchback | Tata Tiago           | Booked      |
|    307 |       207 | KA05LM1290     | Sedan     | Honda Amaze          | Available   |
|    308 |       208 | KA03PQ6643     | SUV       | Maruti Ertiga        | Available   |
+--------+-----------+----------------+-----------+----------------------+-------------+
```

5.5 Contents of BOOKING

The table is shown in two parts so that all columns fit on the page.

```
mysql> SELECT Booking_ID, Customer_ID, Driver_ID, Cab_ID, Booking_Date, Booking_Status FROM BOOKING;
```

```
+------------+-------------+-----------+--------+---------------------+----------------+
| Booking_ID | Customer_ID | Driver_ID | Cab_ID | Booking_Date        | Booking_Status |
+------------+-------------+-----------+--------+---------------------+----------------+
|       5001 |         101 |       205 |    305 | 2026-10-01 08:15:00 | Completed      |
|       5002 |         102 |       201 |    301 | 2026-10-02 09:30:00 | Completed      |
|       5003 |         103 |       203 |    303 | 2026-10-03 18:45:00 | Completed      |
|       5004 |         104 |       207 |    307 | 2026-10-04 07:50:00 | Completed      |
|       5005 |         105 |       208 |    308 | 2026-10-06 21:10:00 | Completed      |
|       5006 |         106 |       201 |    301 | 2026-10-08 13:20:00 | Completed      |
|       5007 |         107 |       205 |    305 | 2026-10-10 10:05:00 | Completed      |
|       5008 |         108 |       203 |    303 | 2026-10-12 17:40:00 | Completed      |
|       5009 |         109 |       207 |    307 | 2026-10-15 06:30:00 | Completed      |
|       5010 |         110 |       208 |    308 | 2026-10-17 19:25:00 | Completed      |
|       5011 |         101 |       203 |    303 | 2026-10-19 11:00:00 | Cancelled      |
|       5012 |         104 |       201 |    301 | 2026-10-20 15:30:00 | Cancelled      |
|       5013 |         102 |       202 |    302 | 2026-10-23 08:00:00 | Confirmed      |
|       5014 |         105 |       206 |    306 | 2026-10-23 09:15:00 | Confirmed      |
|       5015 |         107 |       205 |    305 | 2026-10-24 10:30:00 | Pending        |
+------------+-------------+-----------+--------+---------------------+----------------+
```

```
mysql> SELECT Booking_ID, Pickup_Location, Drop_Location FROM BOOKING;
```

Booking_ID	Pickup_Location	Drop_Location
5001	Alliance University	Electronic City Phase 1
5002	HSR Layout	Kempegowda International Airport
5003	Koramangala	Whitefield
5004	Jayanagar	MG Road
5005	Indiranagar	Marathahalli
5006	Electronic City Phase 2	Silk Board Junction
5007	Alliance University	Bannerghatta Road
5008	BTM Layout	Majestic Bus Stand
5009	Anekal Town	Kempegowda International Airport
5010	Whitefield	Indiranagar
5011	Electronic City Phase 1	Alliance University
5012	MG Road	HSR Layout
5013	Koramangala	Electronic City Phase 1
5014	Marathahalli	Jayanagar
5015	Alliance University	Silk Board Junction

5.6 Contents of TRIP

```
mysql> SELECT * FROM TRIP;
```

Trip_ID	Booking_ID	Start_Time	End_Time	Distance	Fare	Trip_Status
7001	5001	2026-10-01 08:25:00	2026-10-01 09:02:00	18.50	350.00	Completed
7002	5002	2026-10-02 09:40:00	2026-10-02 10:35:00	36.20	780.00	Completed
7003	5003	2026-10-03 19:00:00	2026-10-03 20:05:00	22.80	520.00	Completed
7004	5004	2026-10-04 08:00:00	2026-10-04 08:30:00	9.40	210.00	Completed
7005	5005	2026-10-06 21:20:00	2026-10-06 21:58:00	14.60	290.00	Completed
7006	5006	2026-10-08 13:30:00	2026-10-08 13:52:00	8.10	180.00	Completed
7007	5007	2026-10-10 10:15:00	2026-10-10 11:05:00	21.00	420.00	Completed
7008	5008	2026-10-12 17:55:00	2026-10-12 18:50:00	12.70	260.00	Completed
7009	5009	2026-10-15 06:40:00	2026-10-15 07:55:00	41.50	950.00	Completed
7010	5010	2026-10-17 19:35:00	2026-10-17 20:25:00	17.90	360.00	Completed
7011	5013	2026-10-23 08:10:00	NULL	NULL	NULL	Started
7012	5012	NULL	NULL	0.00	0.00	Cancelled

5.7 Contents of PAYMENT

```
mysql> SELECT * FROM PAYMENT;
```

Payment_ID	Booking_ID	Amount	Payment_Method	Payment_Date	Payment_Status
8001	5001	350.00	UPI	2026-10-01 09:05:00	Successful
8002	5002	780.00	Card	2026-10-02 10:40:00	Successful
8003	5003	520.00	UPI	2026-10-03 20:10:00	Successful
8004	5004	210.00	Cash	2026-10-04 08:35:00	Successful
8005	5005	290.00	UPI	2026-10-06 22:00:00	Successful
8006	5006	180.00	Cash	2026-10-08 13:55:00	Successful
8007	5007	420.00	Card	2026-10-10 11:10:00	Successful
8008	5008	260.00	UPI	2026-10-12 18:55:00	Successful
8009	5009	950.00	Card	2026-10-15 08:00:00	Successful
8010	5010	360.00	UPI	NULL	Pending

5.8 Contents of RATING

```
mysql> SELECT * FROM RATING;
```

```
+-----------+------------+--------+-------------------------------------------------+
| Rating_ID | Booking_ID | Rating | Review                                          |
+-----------+------------+--------+-------------------------------------------------+
|      9001 |       5001 |      5 | Good ride, smooth and punctual.                 |
|      9002 |       5002 |      4 | Reached the airport on time, AC could be cooler. |
|      9003 |       5003 |      5 | Very polite driver and clean cab.               |
|      9004 |       5004 |      3 | Driver took a longer route.                     |
|      9005 |       5005 |      4 | Safe driving and a good experience.             |
|      9006 |       5006 |      5 | Quick pickup and a comfortable ride.            |
|      9007 |       5007 |      4 | Good ride overall.                              |
|      9008 |       5009 |      5 | Early morning pickup was on time.               |
+-----------+------------+--------+-------------------------------------------------+
```

5.9 Contents of DRIVER_LOCATION

```
mysql> SELECT * FROM DRIVER_LOCATION;
```

```
+-------------+-----------+-----------+-----------+---------------------+
| Location_ID | Driver_ID | Latitude  | Longitude | Updated_Time        |
+-------------+-----------+-----------+-----------+---------------------+
|           1 |       201 | 12.916500 | 77.610100 | 2026-10-24 09:00:00 |
|           2 |       201 | 12.925000 | 77.615000 | 2026-10-24 09:30:00 |
|           3 |       202 | 12.845600 | 77.660300 | 2026-10-24 09:00:00 |
|           4 |       202 | 12.850000 | 77.665000 | 2026-10-24 09:30:00 |
|           5 |       203 | 12.935200 | 77.624500 | 2026-10-24 09:00:00 |
|           6 |       203 | 12.940000 | 77.630000 | 2026-10-24 09:30:00 |
|           7 |       204 | 12.969800 | 77.750000 | 2026-10-24 09:00:00 |
|           8 |       204 | 12.970000 | 77.751000 | 2026-10-24 09:30:00 |
|           9 |       205 | 12.710700 | 77.701600 | 2026-10-24 09:00:00 |
|          10 |       205 | 12.716000 | 77.710000 | 2026-10-24 09:30:00 |
|          11 |       206 | 12.956900 | 77.701100 | 2026-10-24 09:00:00 |
|          12 |       206 | 12.960000 | 77.705000 | 2026-10-24 09:30:00 |
|          13 |       207 | 12.925000 | 77.593800 | 2026-10-24 09:00:00 |
|          14 |       207 | 12.930000 | 77.599000 | 2026-10-24 09:30:00 |
|          15 |       208 | 12.978400 | 77.640800 | 2026-10-24 09:00:00 |
|          16 |       208 | 12.981000 | 77.644000 | 2026-10-24 09:30:00 |
+-------------+-----------+-----------+-----------+---------------------+
```

6. Queries Using the Relationships

The following queries join the tables through their foreign keys. They show that the sample data is connected correctly and can be used for reporting.

6.1 Bookings by status

```
mysql> SELECT Booking_Status, COUNT(*) AS Total FROM BOOKING GROUP BY Booking_Status ORDER BY Total DESC;
```

```
+----------------+-------+
| Booking_Status | Total |
+----------------+-------+
| Completed      |    10 |
| Cancelled      |     2 |
| Confirmed      |     2 |
| Pending        |     1 |
+----------------+-------+
```

6.2 Booking details with customer, driver and cab type

```
mysql> SELECT b.Booking_ID, c.Name AS Customer, d.Name AS Driver, cb.Cab_Type, b.Pickup_Location,
    -> b.Drop_Location, b.Booking_Status FROM BOOKING b JOIN CUSTOMER c ON b.Customer_ID =
    -> c.Customer_ID JOIN DRIVER d ON b.Driver_ID = d.Driver_ID JOIN CAB cb ON b.Cab_ID = cb.Cab_ID
    -> ORDER BY b.Booking_ID;
```

Booking_ID	Customer	Driver	Cab_Type	Pickup_Location	Drop_Location	Booking_Status
5001	Rahul Sharma	Venkatesh Rao	SUV	Alliance University	Electronic City Phase 1	Completed
5002	Ananya Iyer	Ramesh Kumar	Sedan	HSR Layout	Kempegowda International Airport	Completed
5003	Karthik Nair	Manjunath Gowda	Hatchback	Koramangala	Whitefield	Completed
5004	Sneha Patil	Prakash Shetty	Sedan	Jayanagar	MG Road	Completed
5005	Arjun Mehta	Mohammed Irfan	SUV	Indiranagar	Marathahalli	Completed
5006	Divya Krishnan	Ramesh Kumar	Sedan	Electronic City Phase 2	Silk Board Junction	Completed
5007	Rohan Das	Venkatesh Rao	SUV	Alliance University	Bannerghatta Road	Completed
5008	Meera Joshi	Manjunath Gowda	Hatchback	BTM Layout	Majestic Bus Stand	Completed
5009	Vikram Singh	Prakash Shetty	Sedan	Anekal Town	Kempegowda International Airport	Completed
5010	Pooja Reddy	Mohammed Irfan	SUV	Whitefield	Indiranagar	Completed
5011	Rahul Sharma	Manjunath Gowda	Hatchback	Electronic City Phase 1	Alliance University	Cancelled
5012	Sneha Patil	Ramesh Kumar	Sedan	MG Road	HSR Layout	Cancelled
5013	Ananya Iyer	Suresh Babu	SUV	Koramangala	Electronic City Phase 1	Confirmed
5014	Arjun Mehta	Anil Reddy	Hatchback	Marathahalli	Jayanagar	Confirmed
5015	Rohan Das	Venkatesh Rao	SUV	Alliance University	Silk Board Junction	Pending

6.3 Successful payments by method

```
mysql> SELECT Payment_Method, COUNT(*) AS Payments, SUM(Amount) AS Total_Amount FROM PAYMENT WHERE
    -> Payment_Status = 'Successful' GROUP BY Payment_Method;
```

Payment_Method	Payments	Total_Amount
Card	3	2150.00
Cash	2	390.00
UPI	4	1420.00

6.4 Average rating of each driver

```
mysql> SELECT d.Driver_ID, d.Name, COUNT(r.Rating_ID) AS Ratings, ROUND(AVG(r.Rating), 2) AS Avg_Rating
    -> FROM DRIVER d JOIN BOOKING b ON b.Driver_ID = d.Driver_ID JOIN RATING r ON r.Booking_ID =
    -> b.Booking_ID GROUP BY d.Driver_ID, d.Name ORDER BY Avg_Rating DESC;
```

Driver_ID	Name	Ratings	Avg_Rating
203	Manjunath Gowda	1	5.00
205	Venkatesh Rao	2	4.50
201	Ramesh Kumar	2	4.50
207	Prakash Shetty	2	4.00
208	Mohammed Irfan	1	4.00

7. Checking Integrity of the Inserted Data

These queries look for rows that break a relationship. Every violation count must be 0.

7.1 Orphan record check

```
mysql> SELECT 'BOOKING without CUSTOMER' AS Check_Name, COUNT(*) AS Violations FROM BOOKING b LEFT JOIN
    -> CUSTOMER c ON b.Customer_ID = c.Customer_ID WHERE c.Customer_ID IS NULL UNION ALL SELECT
    -> 'BOOKING without DRIVER', COUNT(*) FROM BOOKING b LEFT JOIN DRIVER d ON b.Driver_ID =
    -> d.Driver_ID WHERE d.Driver_ID IS NULL UNION ALL SELECT 'BOOKING without CAB', COUNT(*) FROM
    -> BOOKING b LEFT JOIN CAB c ON b.Cab_ID = c.Cab_ID WHERE c.Cab_ID IS NULL UNION ALL SELECT
    -> 'TRIP without BOOKING', COUNT(*) FROM TRIP t LEFT JOIN BOOKING b ON t.Booking_ID =
    -> b.Booking_ID WHERE b.Booking_ID IS NULL UNION ALL SELECT 'PAYMENT without BOOKING', COUNT(*)
    -> FROM PAYMENT p LEFT JOIN BOOKING b ON p.Booking_ID = b.Booking_ID WHERE b.Booking_ID IS NULL
    -> UNION ALL SELECT 'RATING without BOOKING', COUNT(*) FROM RATING r LEFT JOIN BOOKING b ON
    -> r.Booking_ID = b.Booking_ID WHERE b.Booking_ID IS NULL UNION ALL SELECT 'LOCATION without
    -> DRIVER', COUNT(*) FROM DRIVER_LOCATION l LEFT JOIN DRIVER d ON l.Driver_ID = d.Driver_ID
    -> WHERE d.Driver_ID IS NULL;

+--------------------------+------------+
| Check_Name               | Violations |
+--------------------------+------------+
| BOOKING without CUSTOMER |          0 |
| BOOKING without DRIVER   |          0 |
| BOOKING without CAB      |          0 |
| TRIP without BOOKING     |          0 |
| PAYMENT without BOOKING  |          0 |
| RATING without BOOKING   |          0 |
| LOCATION without DRIVER  |          0 |
+--------------------------+------------+
```

7.2 Booking driver and cab consistency

Because of the Driver-Cab 1:1 rule, the driver of a booking must be the driver of its cab.

```
mysql> SELECT 'Booking driver differs from cab driver' AS Check_Name, COUNT(*) AS Violations FROM BOOKING b
    -> JOIN CAB c ON b.Cab_ID = c.Cab_ID WHERE b.Driver_ID <> c.Driver_ID;

+---------------------------------------+------------+
| Check_Name                            | Violations |
+---------------------------------------+------------+
| Booking driver differs from cab driver |          0 |
+---------------------------------------+------------+
```

8. Conclusion

Realistic sample data with 87 records was inserted into all eight tables of the Cab Booking Management System. The data was inserted in the correct order, follows every primary key, foreign key, UNIQUE, CHECK, NOT NULL and DEFAULT rule, and covers different situations such as completed, cancelled, confirmed and pending bookings. The verification queries show that the tables are properly connected and that there are no orphan or inconsistent records.